



Applications of Machine Learning in cricket: A systematic review

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ABSTRACT

Cricket has become a famous team game around the globe, and it is considered the world's second most popular sport (Pathak and Wadhwa, 2016). The plethora of available cricket data and the development of Machine Learning (ML) technology have created a massive demand for cricket data analytics. The applications of ML in the cricket domain have increased dramatically during the last two decades. This study conducts a systematic review of the published research work during the last two decades (2001–2021) on the applications of ML in cricket.

1. Introduction

Cricket is a bat and ball game played in 106 member states of the International Cricket Council (ICC), and it has become a multi-billion dollar business (Kampakis & Thomas, 2015). There are three major formats: Test Cricket, One Day International Cricket (ODI), and the Twenty-Twenty Cricket (T20). Cricket has rapidly become a popular team sport mainly due to the dynamic nature of both ODI and T20 formats (Bandulasiri, Brown, & Wickramasinghe, 2016). Franchise T20 leagues have emerged in various world regions with this increased popularity.

Cricket data analysis has become an integral part of any successful cricket team. The findings of cricket analytics provide a better insight into the players and the game, which is very helpful to people involved in the game, such as current players, technical staff, and managers, and to educate the players of next generation (Morgulev, Azar, & Lidor, 2018; Sarlis & Tjortjis, 2020). Due to the quick evolvement of Cricket, administrators frequently search for innovative ideas to enhance the performance of cricketers to a competitive edge. Intense and stressful situations characterize the sports environment at a competitive level. The duty of the performers in the sport is to handle the pressure and psychological responses to fulfill their potential (Devonport, 2015). The player-performance management is connected with optimizing player performance while minimizing the risk of injury risks involved in various sports (Naglah et al., 2018; Xu & Tang, 2021). In this process, cricket data analysis has an important role to play.

Sports data analytics could not rely solely on traditional statistical procedures due to numerous limitations. Usually, the conventional statistical techniques are based on assumptions about the data, and cricket

data may not satisfy the required assumptions. Furthermore, events related to cricket are not independent events and are influenced by various human factors; therefore, the selection of data analysis techniques has paramount importance (Horvat & Job, 2019; Karlis & Ntzoufras, 2003). In addition, most traditional metrics fail to address modern-day game hypotheses due to a massive volume of related variables. The emergence of Machine learning (ML) has influenced sports data analytics tremendously amid the limitations shown by conventional statistical data analysis techniques.

A branch of artificial intelligence called Machine Learning (ML) is a collection of computer algorithms providing systems the ability to automatically learn and progress with the experience. The main aim of ML techniques is to provide a higher level of automation in the process of knowledge engineering by replacing time-consuming human activities. The abundance of computational power and data has given enormous popularity to ML techniques used in Sports data analytics. The advancement of the ML process and its applications have shown tremendous maturity during the past two decades. Furthermore, sports data analytics have reached a higher level due to the wealth of sport-related data and the development of machine learning techniques. ML techniques, as opposed to conventional computer systems by allowing these systems to learn from data without imposing rules or being explicitly programmed (Grues, 2015).

The introduction of electronic devices to gather sports data has shown a tremendous boost in sports data analytics, with the influence of ML. The power of ML techniques has proliferated due to their impact on the devices used to collect data (Jamil, Iqbal, Ahmad, & Kim, 2020), extract information from the devices (Morris, Mundt,

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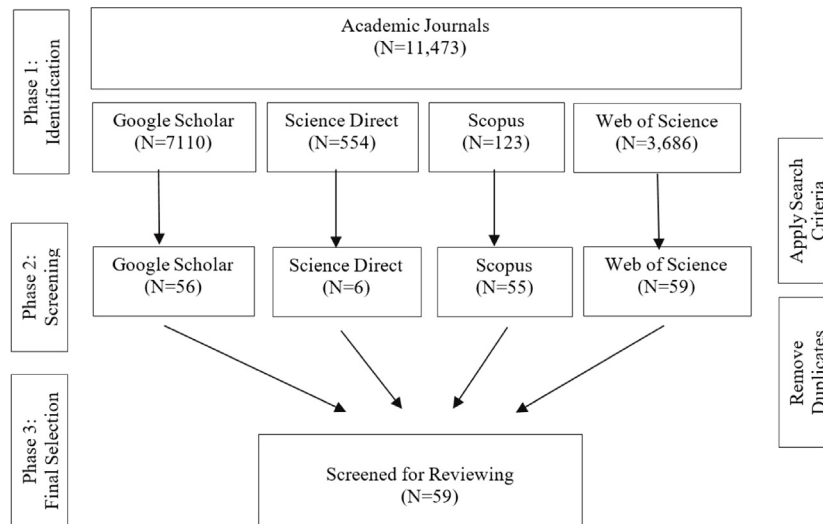


Fig. 1. Reviewing process.

Goldacre, & Jacqueline, 2020; Weir, Alderson, Smailes, Elliott, & Donnelly, 2019), and the processing the information gathered through the devices to enhance the understanding of the ultimate users (Ahmadi et al., 2014; Rommers et al., 2020). Most of these electronic devices come as wearable devices. The amalgamation of these wearables and ML algorithms has opened the gate for developing AI expert systems used in sports (Acikmese, Ustundag, & Golubovic, 2017). These ML-based applications quickly grasped sports data analytics to construct models and predict future outcomes with existing sports data to deliver accurate predictions to make better decisions in sports. Applications of ML quickly grasped sports data analytics to construct models and predict future outcomes with existing sports data to deliver accurate predictions to make better decisions in sports. The development of these ML systems requires domain-specific knowledge. In order to optimize ML algorithms, it uses a process called feature engineering. The purpose of this is to extract features from the raw data with the help of domain knowledge to optimize the ML algorithms. Though sports data are abundant, applying ML for big data is thought-provoking as it demands expert knowledge of the domain, used learning algorithms, and software engineering (Koseler & Stephan, 2017).

Systematic reviews help collect empirical evidence about the research's continuous growth using ML in the cricket domain. Furthermore, according to the author's knowledge, there is no such systematic review of two decades of research findings combining cricket and ML. Therefore, this work intends to fill the above vacuum in the literature, and the findings of this study will be beneficial for the players, coaches, and sports administrators. Finally, this effort will help researchers to have a concise overview of the existing research areas in cricket and identify research gaps for their future research work.

2. Methodology

This manuscript systematically studied how machine learning techniques have been applied in cricket in contemporary literature. As a reference, standardized guidelines proposed by Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA; Moher, Liberati, Tetzlaff, Altman, & Prisma Group*, 2009) for systematic reviews were followed. The entire review protocol comprises three phases: identification, screening, and final selection. Fig. 1 outlines these three phases of the review protocol.

First we identified all published work using four major databases, namely, Google Scholar, Science Direct, Scopus, Web of Science. This search using broader research domains was conducted in February

2022. The electronic search was conducted using the keywords "Applications of Machine Learning in Cricket". All published research work satisfying the above searching information was preliminarily identified.

Secondly, we selected the most relevant publications based on the following criteria.

- Studies directly related to the game of cricket, including but not limited to the player performance, the game's outcome, team performance, prediction of the score, pitch related, cricket commentary, and videoing the cricket games were considered.
- Only the human game of cricket is considered by discarding the robot or electronic versions of the game.
- Studies utilizing at least one ML technique were selected.
- Studies published in a peer-reviewed journal from 2001 to 2021 were considered.
- Both conference and journal versions of publications were included.

Finally, duplicates were removed for the final selection. After removing the duplicates from each database, only 59 publications remained for the final selection.

3. Results

This section aims to present and discuss the findings of this systematic review. First, a discussion is conducted to identify the research areas in cricket where the applications of ML can be seen. Then, some descriptive statistics are given about the reviewed publications. Secondly, in the research question section, all the selected articles are reviewed to identify which studies have addressed the research questions related to which type of feature extractions, ML technique, and accuracy estimating technique were used. The following subsection reviews the data sets and the number of attributes used in each study. Then, findings of the used data reduction and feature extraction techniques are presented. After that, a review of the frequently used ML techniques is presented. The final section aims to quantify the accuracy of the used ML techniques.

3.1. Study areas in cricket

According to findings, studies involving cricket and ML techniques have increased exponentially since 2001. Among these publications, the focus on the ODI game has increased significantly compared to the other formats. Fig. 4 displays the number of studies conducted using ML techniques from 2001 to 2021 related to each format of cricket.

Table 1
Research areas in cricket.

Research area	% of studies
Game outcome prediction	35
Player's performance classification	16
Batting style/Stroke classification	9
Bowling action/performance classification	9
Other	9
Umpire's decision/gestures	6
Score prediction	5
Cricket commentary/media	4
Pitch behavior prediction	4
Team selection/performance	3

Further investigations show that out of all areas in cricket, about 35% of the studies have focused on predicting the game's outcome. About 16% of the studies have been conducted studies in player performance classification. More information about the researched areas is listed in Table 1. The following sections summarize the contribution of reviewed articles corresponding to each section.

- **Game outcome prediction:** With the increased popularity and the commercialization of the game, outcome prediction of a cricket game has become of the utmost importance. In this direction, researchers use various performance indicators representing various aspects of the game and a wide range of ML techniques to predict the game's outcome. Batsman, bowler, and fielder attributes (Hasanika, Dilhara, Liyanage, Bandaranayake, & Deegalla, 2021; Karthik et al., 2021; Modani, Kilaru, Kaur, Sinha, & Khetan, 2020) are the most commonly used performance measures to use in the prediction. In addition, other attributes such as home game advantage (Kaluvarachchi & Aparna, 2010; Kumar, Santhadevi, & Barnabas, 2019), the outcome of the toss (Pathak & Wadhwa, 2016), and the behavior of the pitch (Tekade, Markad, Amage, & Natekar, 2020) have been utilized in the prediction. More work in outcome prediction can be seen in Basit et al. (2020), Hatharasinghe and Poravi (2019), Shakil, Abdullah, Momen, and Mohammed (2020), Vistro, Rasheed, and David (2019).
- **Player's performance classification:** Though batting, bowling, and fielding are considered the three main departments of the game, batting and bowling are the most popular statistics used to quantify players' performance (Wickramasinghe, 2014b). Though batting, bowling, and fielding are considered the three central departments of the game, batting and bowling are the most popular statistics used to quantify players' performance (Wickramasinghe, 2014b). Individual players' skills have a good impact on the game's outcome. Therefore, assessing individual skill levels is imperative in many ways to the game; for this, ML techniques can be used together with player-level parameters (Wickramasinghe, 2020a). Numerous such instances can be seen as Classifying players into various skill levels (Aburas, Mehtab, & Mehtab, 2018; Manage, Kafle, & Wijekularathna, 2020; Wickramasinghe, 2020b), evaluating or ranking them (Ahmad et al., 2021; Premkumar, Chakrabarty, & Chowdhury, 2020), identifying or predicting the player performance level (Anik, Yeaser, Hossain, & Chakrabarty, 2018; Mody, Malathi, & Jayaseeli, 2021; Rupai, Mukta, & Islam, 2020), and predicting the best batsmen/bowler (Rani et al., 2020) or the rising stars (Ahmad et al., 2017). With the help of identifying the best performers, a strong team can be formed as the performance of the individual players have a direct impact on the rank of the team (Wickramasinghe, 2014a). Evidence of the use of ML techniques to select the best team can be seen in Ishi and Patil (2020), Mahbub, Miah, Islam, Sorna, Hossain, and Biswas (2021).
- **Batting style and bowling action classification:** Cricket is considered a batsmen's game requiring the ability to make shot selection and execution. ML techniques coupled with body-worn internal

measurement units (IMUs) have been used to identify and assess shot making skills of the batsmen (Dias, Mitchell, & Harland, 2020). Furthermore, ML and the wearable devices can used to study the bowler's characteristics such as bowling action (Salman, Qaisar, & Qamar, 2017), bowling volume, ball releasing speed, and identifying the intensity zones (McGrath, Neville, Stewart, Clinning, & Cronin, 2021; McGrath, Neville, Stewart, Clinning, Thomas, & Cronin, 2021; McGrath, Neville, Stewart, & Cronin, 2019; Ranaweera & Silva, 2019). Moreover, accurate evaluation of the bowler's workload can be monitored and predicted with the use of ML, and it will, in turn, minimize the possible future player injuries (Jowitt, Durussel, Brandon, & King, 2020).

- **Other areas:** In addition to the above areas, ML has been used in media coverage of cricket, umpire-related activity prediction, pitch behavior predication, and score prediction.

Most international cricketers are actively engaging with their fans through social media platforms. This review reveals the use numerous media platforms including social media together with ML in predicting cricket related information (Dubey, Suri, & Gupta, 2021; Mustafa, Nawaz, Lali, Zia, & Mehmood, 2017; Wickramasinghe & Yapa, 2018; Zakzouk & Mathkour, 2012). Video analytics in cricket has evolved into an exciting area with the development of ML technology. Applications of ML techniques in cricket video studies can be seen in Gupta and Muthiah (2020), Sen, Deb, Dhar, and Koshiba (2021), and Kumar, Kumar, and Kumar (2018). With the enhancement of the game's competitiveness, at times, umpires are under some pressure as their decision can hugely impact the outcome of the game. With technology coupled with ML techniques, systems can be developed to mitigate human errors from umpires. Such efforts can be seen in Aftab, Hussain, Waleed, Ashfaq, and Umair (2019), Iyer, Bala, Sohan, Dharmesh, and Raman (2020), Mustafa et al. (2017), and Samaraweera, Premaratne, and Dharmaratne (2020).

According to the reviewed articles, score prediction (Kamble, 2021; Singh, Singla, & Bhatia, 2015; Srinivas, Bhat, & Revanasiddappa, 2021) is another area of study using the ML techniques. Furthermore, a significant number of studies have been conducted to various aspects of the game such as Duckworth-Lewis (Abbas & Haider, 2019), injuries (Dias et al., 2020; Gupta & Muthiah, 2020), cricket data analytic (Jayalath, 2018; Kamble, 2021; Kapadia, Abdel-Jaber, Thabtah, & Hadi, 2020; Parameswaran, 2013; Rahman, Shamim, & Ismail, 2018; Raja, Manasa, Reddy, & Sundari, 2021; Shahjalal, Ahmad, Rayan, & Alam, 2017; Srinivas et al., 2021), and predicting the behavior of the pitch (Kanhaiya, Gupta, & Sharma, 2019).

3.2. Descriptive statistics about the studies

This section summarizes the descriptive statistics of the reviewed articles. According to the findings, 56% of the studies were published in peer-reviewed journals, and the remainder were published in conferences. Fig. 2 illustrates further information about the type of publication and the published year. Since 2010, there has been a rapid increment in the number of studies in cricket using ML. Only 10% of the publication can be seen from 2010-to 2016. After that, for each year from 2017 to 2021, there are 8%, 12%, 14%, 30%, and 25% publications were recorded. More information can be seen in Fig. 3.

3.3. Research questions

This section focuses on three main research questions as listed below. We identify whether each reviewed study has focused on each of the following identified research questions, and a summary is illustrated in Table 2.

Table 2
Study and the addressed research questions.

S/N	Reference	RQ1	RQ2	RQ3	S/N	Reference	RQ1	RQ2	RQ3
1.	Abbas and Haider (2019)	N	Y	N	31.	McGrath, Neville, Stewart, Clinning, Thomas et al. (2021)	Y	N	Y
2.	Aburas et al. (2018)	N	N	N	32.	Modani et al. (2020)	Y	Y	Y
3.	Aftab et al. (2019)	Y	Y	Y	33.	Mody et al. (2021)	Y	Y	Y
4.	Ahmad et al. (2017)	Y	Y	Y	34.	Mustafa et al. (2017)	N	Y	Y
5.	Ahmad et al. (2021)	Y	Y	Y	35.	Nandyal and Kattimani (2021)	Y	N	Y
6.	Anik et al. (2018)	Y	Y	Y	36.	Panda, Sathya, Mishra, and Satpathy (2019)	N	Y	N
7.	Awan et al. (2021)	N	N	Y	37.	Parameswaran (2013)	N	N	Y
8.	Basit et al. (2020)	N	Y	Y	38.	Pathak and Wadhwa (2016)	N	Y	Y
9.	Deval, Hamid, and Goel (2021)	N	Y	N	39.	Premkumar et al. (2020)	N	N	N
10.	Dias et al. (2020)	Y	Y	Y	40.	Rahman et al. (2018)	N	Y	N
11.	Dubey et al. (2021)	N	Y	N	41.	Raja et al. (2021)	N	Y	N
12.	Goggins et al. (2021)	N	Y	Y	42.	Ranaweera and Silva (2019)	Y	N	Y
13.	Gupta and Muthiah (2020)	Y	Y	N	43.	Rani et al. (2020)	N	Y	N
14.	Hasanika et al. (2021)	Y	Y	N	44.	Rupai et al. (2020)	Y	Y	Y
15.	Hatharasinghe and Poravi (2019)	Y	Y	N	45.	Salman et al. (2017)	Y	Y	Y
16.	Ishi and Patil (2020)	N	Y	N	46.	Samaraweera et al. (2020)	Y	Y	Y
17.	Iyer et al. (2020)	N	Y	Y	47.	Sen et al. (2021)	Y	Y	Y
18.	Jowitt et al. (2020)	Y	N	N	48.	Shakil et al. (2020)	Y	Y	Y
19.	Jayalath (2018)	N	Y	N	49.	Shahjalal et al. (2017)	N	N	Y
20.	Kamble (2021)	N	Y	N	50.	Singh et al. (2015)	N	Y	N
21.	Kaluvarachchi and Aparna (2010)	N	Y	Y	51.	Somaskandhan, Wijesinghe, Wijegunawardana, Bandaranayake, and Deegalla (2017)	Y	Y	N
22.	Kanhaiya et al. (2019)	N	N	N	52.	Srinivas et al. (2021)	Y	Y	N
23.	Kapadia et al. (2020)	Y	Y	Y	53.	Tekade et al. (2020)	Y	Y	N
24.	Karthik et al. (2021)	N	Y	Y	54.	Tyagi, Kumari, Makkena, Mishra, and Pendyala (2020)	Y	Y	Y
25.	Kumar et al. (2018)	N	Y	Y	55.	Vistro et al. (2019)	Y	Y	N
26.	Kumar et al. (2019)	Y	N	N	56.	Wickramasinghe and Yapa (2018)	Y	Y	Y
27.	Mahbub et al. (2021)	N	Y	N	57.	Wickramasinghe (2020a)	N	N	Y
28.	Manage et al. (2020)	N	Y	Y	58.	Wickramasinghe (2020b)	N	Y	N
29.	McGrath et al. (2019)	N	Y	Y	59.	Zakzouk and Mathkour (2012)	Y	Y	Y
30.	McGrath, Neville, Stewart, Clinning et al. (2021)	Y	Y	Y					

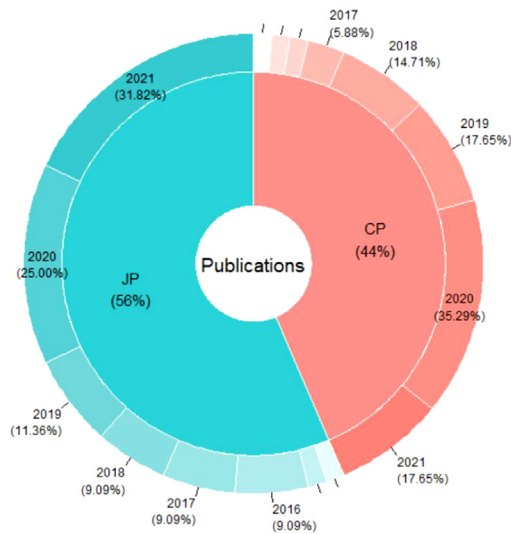


Fig. 2. Publication type (CP-Conference Paper; JP-Journal Paper) vs year.

- Research Question #1, RQ1: Which standard feature extraction techniques have been used and discussed? RQ1 aims to identify the type of feature extraction technique techniques that have been used in cricket data analysis.
- Research Question #2, RQ2: Which ML techniques have been used? RQ2 aims to identify ML techniques that have been used in cricket-related research activities.
- Research Question #3, RQ3: How to estimate the accuracy of the used ML model(s)? RQ3 deals with identifying different ways to quantify the overall accuracy of the used ML model(s).

Table 3
Descriptive statistics of the used data set.

Format	Mean	SD	Min	Q1	Q2	Q3	Max
First Class	861.8	774.3	44	281.0	909.5	1490.2	1584
ODI	27239.6	93191.0	128	385.5	1301.5	5603.8	350899
T20	537502.0	1400273.0	140	994.0	16720.0	123514.0	4000000
Test	1666.0	1856.2	354	1010.0	1666.0	2323.0	2979

Using the selected publications to review, studies that have focused on the above research questions were identified. Table 2 indicates whether the study considered (Y-Yes, N-No) each of the research questions (RQ1, RQ2, and RQ3).

3.4. Cricket data and the features used

The quality and the accuracy of the study directly depend on using a representative sample. In addition, bias in the inference can be minimized using a larger sample. Furthermore, selecting a good representation of features also directly impacts the quality of the outcome of the ML-based data analysis. Therefore, selecting a good data set is foremost essential. This section summarizes the sizes of data sets used in the selected studies. According to the reviewed studies, the average number of instances of data used in research in the T20 format is 537,502, which is the highest compared to the other formats of cricket. In ODI-related studies, approximately an average of 27,240 instances of records have been used. For the Test and First-Class cricket, these are approximately 1666 and 862, respectively (see Table 3).

Feature selection is an integral part of when using an ML technique as it directly impacts the accuracy of the ML technique. Due to the popularity of the game and the advancement in technology, cricket data has become easy to access. Further, the feature set of the cricket game expanded accordingly. According to the findings, studies conducted using First-Class games have used 144 features on average, which

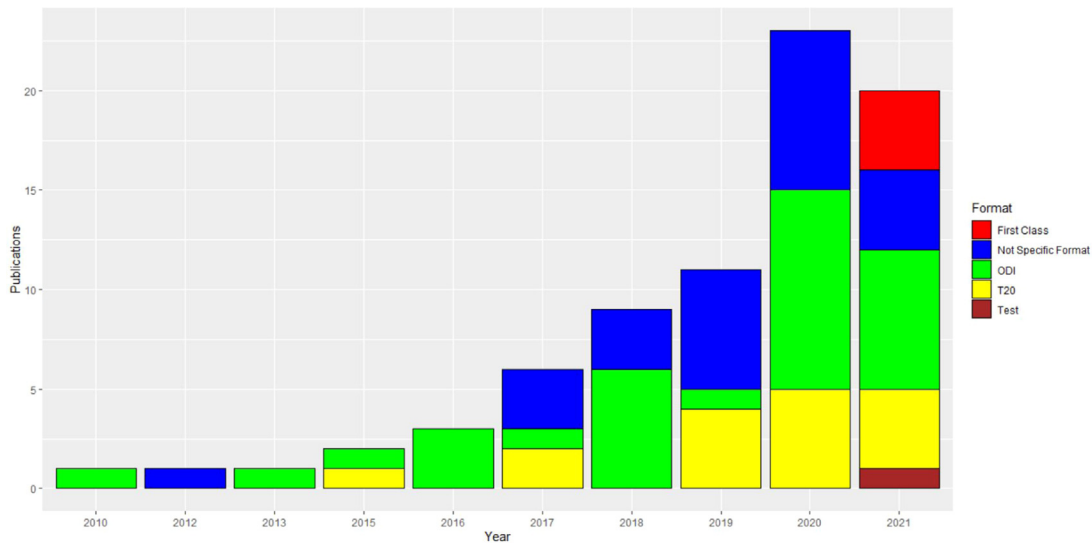


Fig. 3. Format of cricket and the year of study.

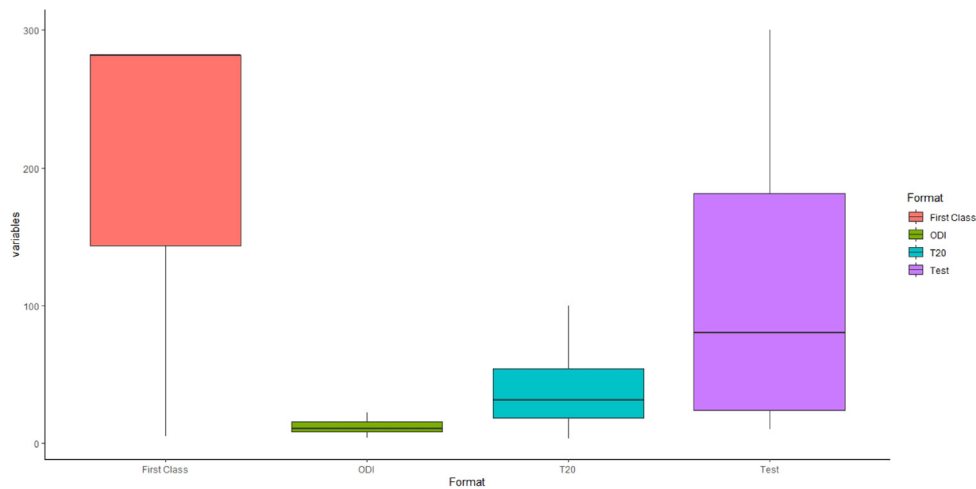


Fig. 4. Number of features used in studies.

is the highest compared to the other game formats. Secondly, the studies conducted using Test cricket have used 115 features on average, while for ODI and T-20, the average number of features is 40 and 12, respectively. Fig. 2 illustrates the distribution of the number of features used in each of the formats of the game.

3.5. Used data reduction and feature extraction techniques

Feature selection is a critical step in ML. The focus of the feature selection approach is to choose a subset of variables from the input data to describe the input data efficiently while reducing effects from noise or irrelevant variables and still providing good prediction results (Guyon & Elisseeff, 2003). Based on the reviewed studies, the following techniques were able to identify as the primary feature selection technique.

- Chi-Square
- Correlation Based Feature Selection
- Graphics Based Feature Selection

- Principal Component Analysis
- Recursive Feature Elimination
- Regression

Among the above list, Correlation based feature selection techniques, Recursive Elimination, and the Chi-Square techniques were the most frequently used.

3.6. Frequently used ML techniques

This section studies the frequently used ML techniques in cricket. According to the reviewed studies, the following ML techniques have been used as the popular ML techniques to analyze cricket data.

- Regression
- Naïve Bayes (NB)
- K-Means
- Random Forest (RF)
- Decision Trees (DT)
- k th Nearest Neighbor (kNN)

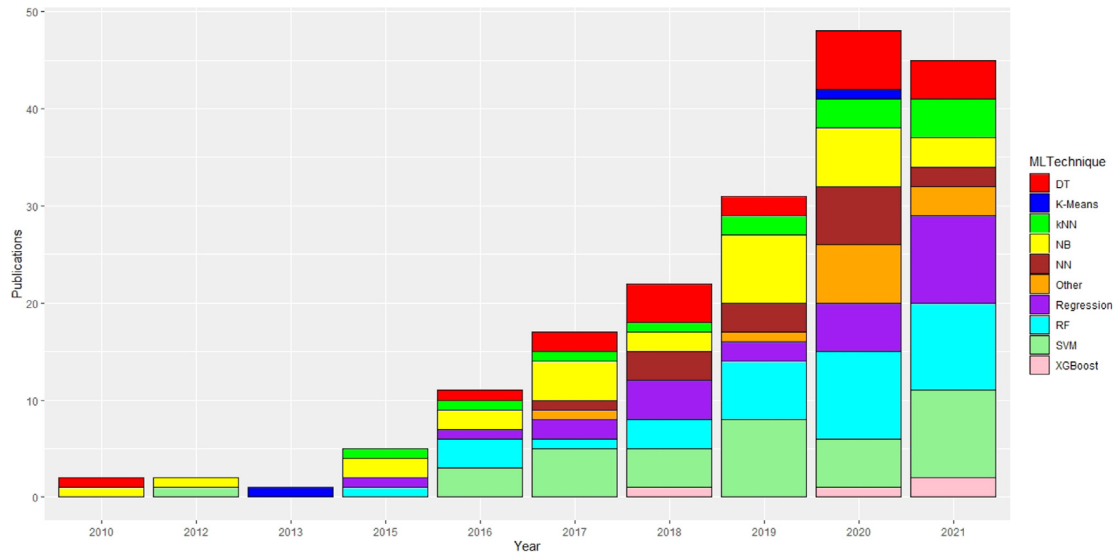


Fig. 5. Year, ML technique and publications.

Table 4

Used ML techniques.

ML technique	Applications	% of studies
SVM	Umpire's gesture modeling	45
	Fast bowler's action modeling	
	Game's outcome prediction	
	Player-performances classification	
RF	Predicting the winner of the tournament	42
	Predicting the winner of the game	
	Classification of all-rounders	
NB	Predicting the winner of the game	36
	Prediction the Batting and bowling performance	
	Pitch behavior prediction	
Regression	Identification of game's influential factors	31
	Innings' score prediction	
DT	Identification of game's influential factors	26
	Winner prediction	
NN	Batting and bowling performance prediction	21
	Game outcome prediction	
	Cricket-shot classification	
kNN	Game's score prediction	17
	Classification of all-rounders	
Other	Player-performances classification	14
	Predicting the player-ranking	
XGBoot	Cricket and social media	5
	Bowler's workload prediction	
K-Means	Batsmen classification	3

- Artificial Neural Networks (ANN)
- Support Vector Machine (SVM)
- XGBoot

Predicting the duration of a game Player performance modeling Among the above-listed ML techniques used in the reviewed papers, SVM (45%) is the most frequently used technique. RF (42%) has been the second most frequently used ML technique, followed by NB (36%), the third most frequently used ML technique. According to the findings, 45% reviewed studies have used SVM, and 26% and 24% of them were published in 2019 and 2021. The second most used ML technique is RF. There are 42% published research have used RF, and 29% of them have been used in 2020 and 2021. The third popular ML technique used in the published works from 2010 to 2021 was NB . Out of all the published works, 36% studies have used NB. Of them, 26% of the work was published in 2019, and 22% were used in 2020. More details, including some of the research areas above ML techniques, have been used can be seen in Table 4. According to the findings, many studies using ML techniques with cricket data have increased during the last

Table 5

Used accuracy testing techniques.

ML accuracy technique	% of studies
Confusion matrix based	57.7
F-score	15.5
RMSE	6.2
ROC	9.3
Other techniques	3.1
Cohen's kappa statistic	2.1
MAE	2.1
MCC	2.1
R^2 -statistic	2.1

two decades. Fig. 5 shows the distribution of publications using the above-stated ML techniques from 2001 to 2021.

3.7. Quantifying the accuracy of ML technique

Since all ML models are data-driven, it is essential to evaluate the accuracy of the ML model. Perhaps this is the most important element of ML models. The reviewed studies show that the Confusion matrix-based accuracy measures are the most popular (about 58%) method to test the accuracy of the ML model. The rest of the popular techniques are F-Score, Receiver Operating Characteristic (ROC) curve (about 9%), and Root Mean Square Error (RMSE) (about 6%). Table 5 as stated before, the main accuracy testing technique is the confusion matrix-based technique. Accuracy, Balance Accuracy, Custom Accuracy, Precision, Recall, Sensitivity, and Specificity are some confusion matrix-based techniques researchers have used.

3.8. Some of the addressed problems

The aim of this section is to have a glance at some of the studies conducted to solve existing issues in cricket with the help of ML techniques.

The team selection process is considered tedious, and picking the most suitable eleven players for an upcoming game involves lots of brainstorming sessions. Mahbub et al. (2021) addressed this problem through the player strength-weakness-ranking mechanism. After collecting batsman and bowler characteristics, they constructed two novel performance indicators for batsman and bowler as the base for classifying players. They trained several ML models, SVM, NB, and RF, and the SVM model outperformed the rest.

Identification of the legality of bowling action is a daunting task due to the complicated bio-mechanical movements of the bowler's arm, and the on-field umpire is unable to observe the bowler's action accurately. [Salman et al. \(2017\)](#) used inertial measurement units (IMU) and ML techniques to classify the bowler's bowling action about the delivery's legality. In the first stage, the authors collected data about various bowling actions. In the profiling stage, they identified critical events of the legal bowling action. Next, features gathered using inertial sensors fixed on the bowler's arm were extracted. Finally, several ML techniques, SVM, kNN, NB, RF, and ANN, were trained to classify the bowling action by achieving higher classification accuracy.

[Wickramasinghe \(2020a\)](#) proposed an ML approach to predict the winner of an ODI game. Instead of the conventional regression approach, the author used the ML technique due to the lack of instances required to run a regression analysis. After collecting a dataset representing batting, bowling, and team-related features, univariate, recursive elimination, and principle component analysis (PCA) were used to select the most influential features for the prediction. An NB model was trained using the finalized data to achieve a higher level of prediction accuracy. In another study, [Wickramasinghe \(2020b\)](#) used NB, kNN, and RF models to predict all-rounders in the game accurately. These models were trained on a dataset representing various aspects of bowling, batting, and fielding performance indicators, and RF turned out to be the best predictor.

Making the third innings declaration is a tricky decision due to the various factors needed to consider to take such a decision. [Deval et al. \(2021\)](#) studied the declaration of the third innings of a test cricket match. In particular, they constructed a decision support system to predict the result of a test match at different phases of the match so that the optimal time to declare the inning can be predicted. After gathering a large collection of variables of the game's aspects, they constructed three models to predict the game's outcome before it started, after the second innings, and after the third innings. Though they used several ML techniques such as ANN, Regression, RF, SVM, and XGboost, the SVM appeared to be the most effective one.

Distinguishing the gestures of the umpire is considered a tiresome process in cricket videos. [Nandyal and Kattimani \(2021\)](#) used SVM to classify umpire action and non-action gestures, which is vital in video summarization and automatic sports highlights generation. They developed a system for the identification of umpire action and non-action gestures. In this process, the authors trained an SVM model using two datasets representing the umpire's performing and non-performing activities relating to specific occasions, such as showing boundaries, illegal bowling, and player dismissals, to classify each action as an Umpire Action or Non-Action.

Fast bowlers are more prone to injuries compared to slow bowlers and batters. Unfortunately, no concrete criteria exist to quantify the maximum tolerated workload of bowlers. [McGrath, Neville, Stewart, Clinning et al. \(2021\)](#) used IMUs and ML models to quantify bowling volume (BV), ball release speed (BRS), and perceived intensity zone (PIZ). Using IMUs, data are collected from bowlers in a cross-sectional study setting. Though they used multiple ML techniques, XGBoost turned out to be the most accurate in predicting the BV, and both RF and XGBoost showed the best performances with BRS.

4. Discussion and conclusion

Based on the conducted systematic review, it is evident that the volume of research in cricket using ML technology has been increasing since 2001. This tendency is a positive sign for the game, as the findings of these research outcomes will eventually help the game's development. Improved ML techniques enable the investigation of various cricket aspects that were inconceivable to conduct before using the existing procedures. Furthermore, the amalgamation of ML techniques and electronic wearable devices in cricket research provides numerous benefits. Research findings allow cricketers to deliver accelerated performances by lowering injury risks.

This study reveals that most cricket-player-related studies relied on a conventional set of features associated with batting and bowling. With the introduction of technology, the game evolved by adding novel features related to the game. Therefore in future research, the time has come to search for these novel features while using the frequently used attributes of the game. Moreover, with the introduction of novel rules in cricket, such as fielding restrictions, the number of balls used, the type of balls used, and player substitution, the nature of the current game deviated from the past. Therefore, if we gather cricket data that spans several decades, there is an issue with the compatibility of the data, which can result in inconsistent outcomes.

Cricket comprises three disciplines: batting, bowling, and fielding, which are equally crucial for the game's outcome. According to the reviewed articles, the number of studies has not been distributed evenly among the above three departments. Even within studies on bowlers, the focus has been mainly on fast bowlers. It is rare to see any research conducted using medium pacers and spinners. When considering studies related to player performance, one of the factors that could impact their performance is the type of cricket ball the game uses. There are several brands of cricket balls, namely, Kookaburra, Dukes, and SG. Unfortunately, the literature does not indicate any significant research work in this direction to study the impact of the type of cricket ball on player performance. Furthermore, there is a significant number of studies related to batting, though there are hardly any studies related to fielders. However, fielding plays an essential role in the game's outcome, yet it is surprising to see the lack of studies conducted regarding the fielding aspect.

From an ML point of view, some reviewed studies have not revealed vital information related to the ML process. The nature of the data (size, number of attributes, correlation structure, and sparsity), the data pre-processing technique, feature selection technique, the accuracy quantifying technique, and the level of accuracy the ML technique reaches are some of them.

The reviewed articles showcased the application of various ML techniques in different research areas in cricket. It is hard to recommend a single ML technique for a given application in cricket, as the success of the ML technique depends on various aspects. Based on these articles, there has been an increment in the use of kNN, Regression, and RF compared to the other ML techniques during the last two decades. Other ML techniques such as neural networks and Naive Bayes do not show increased popularity, while SVM still shows its popularity in cricket applications. One of the significant observations was the lack of the use of some popular ML techniques, such as deep learning reinforcement learning and natural language processing. It is expected to see future research using the latest ML techniques such as automated machine learning, multi-modal learning, multi-objective models, and tiny ML to address more sophisticated issues in cricket.

We identify several limitations of this systematic review. First, the study considered only four databases. Furthermore, due to limited resources, this study could not consider another version of cricket, called T-10. This review identified various research activities conducted in cricket using ML and other areas that require researchers' attention. As cricket is a rapidly changing game, more research studies are required in the future. In this direction, ML will play a significant contribution which will eventually help to develop the game further.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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